

Consumer Shopping Behavior Analysis

A Data-Driven Retail Strategy Report

Dataset: 3,900 Customer Records | 18 Features

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Prepared for: Retail Management & Marketing Strategy Teams

Tools Used: Python (pandas) · PostgreSQL · Power BI

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1. Executive Summary

This report summarizes a structured analysis of retail customer shopping behavior using a dataset of 3,900 transactions. The goal was to move past generic reporting and get to something actually useful: which customers are buying what, when, and why — and what the business can do with that information.

The analysis was carried out in three stages. First, the raw data was cleaned and transformed in Python using pandas. Then the cleaned dataset was pushed into a PostgreSQL database for structured querying. Finally, the findings were prepared for visualization in Power BI.

3,900	18	\$59.76	3.75 / 5	27%
Total Records	Features	Avg. Purchase	Avg. Rating	Subscribed

Bottom Line: Most shoppers are occasional buyers spread evenly across seasons. Discounts drive about 43% of purchases. The loyalty subscriber base is small (27%) but likely more valuable per customer. The biggest opportunity is converting infrequent buyers into regulars — which the data suggests is possible through targeted promotions and better subscription incentives.

2. Dataset Overview

The dataset contains 3,900 rows and 18 columns. Each row represents a single customer transaction and includes demographic details, product information, purchase context, and behavioral signals.

Column	Description
customer_id	Unique customer identifier
age	Customer age (18–70)
gender	Male / Female
item_purchased	Specific item bought (25 unique items)
category	Clothing, Accessories, Footwear, Outerwear
purchase_amount	Transaction value in USD (\$20–\$100)
location	US state (50 states represented)
size	S / M / L / XL
color	25 color options
season	Spring, Summer, Fall, Winter

review_rating	Customer rating (2.5–5.0)
subscription_status	Whether customer holds an active subscription
shipping_type	6 shipping methods
discount_applied	Whether a discount was applied
previous_purchases	Count of past transactions
payment_method	PayPal, Credit Card, Cash, Debit, Venmo, Bank Transfer
frequency_of_purchases	Self-reported purchase frequency

Table 1: Dataset feature summary

3. Data Preparation & Methodology

3.1 Missing Value Treatment

Only one column had missing data: *review_rating*, with 37 null entries (less than 1% of the dataset). A simple global median fill would have worked, but it felt sloppy — a customer who bought outerwear and one who bought accessories probably rate things differently. Instead, the nulls were replaced using the category-wise median, which keeps the imputed values contextually grounded.

The fix in code:

```
df['review_rating'] = df.groupby('category')['review_rating'].transform(lambda x:
x.fillna(x.median()))
```

3.2 Column Normalization

All column names were converted to lowercase snake_case to avoid errors during SQL querying and pandas operations. The column *purchase_amount_usd* was renamed to *purchase_amount* for cleaner syntax.

3.3 Redundancy Check

Two columns — *discount_applied* and *promo_code_used* — turned out to be perfectly identical across all 3,900 rows. Keeping both would have been misleading, suggesting two independent signals where there was only one. The *promo_code_used* column was dropped.

4. Key Findings

4.1 Demographics & Purchasing Power

The customer base skews heavily male — 2,652 men vs. 1,248 women. The average age is 44, with customers ranging from 18 to 70. Despite the gender gap in headcount, women spend slightly more on average (\$60.25 vs. \$59.54), so the gender imbalance in the data likely reflects sampling rather than a true behavioral difference.

Total Customers	3,900	Age Range	18 – 70 years
Male Customers	2,652 (68%)	Average Age	44 years
Female Customers	1,248 (32%)	Median Age	44 years
Avg. Spend (Male)	\$59.54	Avg. Spend (Female)	\$60.25

Table 2: Demographic breakdown

4.2 Category & Seasonal Trends

Clothing is the dominant category, accounting for 44.5% of all transactions. Accessories come in second at 31.8%, followed by Footwear (15.4%) and Outerwear (8.3%). Interestingly, average spend per transaction is almost identical across all four categories, hovering around \$59–60. This means the category gap is really about volume, not ticket size.

Category	Transactions	Share	Avg. Purchase
Clothing	1,737	44.5%	\$60.03
Accessories	1,240	31.8%	\$59.84
Footwear	599	15.4%	\$60.26
Outerwear	324	8.3%	\$57.17

Table 3: Sales by category

Purchase volume is evenly distributed across seasons — Spring (999), Fall (975), Winter (971), Summer (955). There are no dramatic seasonal spikes, which is unusual. This could mean the business is well-diversified by product, or it could mean seasonal campaigns aren't being run aggressively enough to create peaks.

4.3 Discount & Promotional Behavior

43% of purchases involved a discount. That's a significant share. The data shows that *discount_applied* and *promo_code_used* were perfectly correlated — every discounted purchase used a promo code and vice versa. This suggests the company runs a single, centralized promo strategy rather than multiple independent discount channels.

Watch-out: When 43% of purchases require a promo code, it raises a question about margin. If shoppers are waiting for discounts before buying, that's a pricing problem, not a loyalty win.

4.4 Payment Methods & Shipping Preferences

Payment preferences are remarkably evenly spread across six methods: PayPal (17.4%), Credit Card (17.2%), Cash (17.2%), Debit Card (16.3%), Venmo (16.3%), and Bank Transfer (15.7%). No single payment method dominates, which has real infrastructure implications — the business needs to maintain all six reliably.

Payment Method	Count	Share
PayPal	677	17.4%
Credit Card	671	17.2%
Cash	670	17.2%
Debit Card	636	16.3%
Venmo	634	16.3%
Bank Transfer	612	15.7%

Table 4: Payment method breakdown

Shipping is similarly distributed. Free Shipping leads slightly (675 orders), but the margin between methods is narrow — from 627 (2-Day Shipping) to 675. No shipping method is clearly preferred, which suggests customers pick based on context rather than habit.

4.5 Customer Loyalty & Subscription

Only 27% of customers hold an active subscription. The remaining 73% are non-subscribers. The average number of previous purchases across all customers is 25, with a max of 50 — so there's a reasonably loyal customer base in the data, but most of them aren't formalizing that loyalty through a subscription.

Purchase frequency is spread across seven categories, with 'Every 3 Months' being the most common (584 customers). Weekly buyers are the rarest segment (539 customers). The takeaway: most customers buy occasionally, not regularly. Converting even a fraction of quarterly buyers to monthly would have a meaningful impact on revenue.

Purchase Frequency	Count	Share
Every 3 Months	584	15.0%
Annually	572	14.7%
Quarterly	563	14.4%
Monthly	553	14.2%
Bi-Weekly	547	14.0%
Fortnightly	542	13.9%
Weekly	539	13.8%

Table 5: Purchase frequency distribution

5. Feature Engineering

Two new columns were added to the dataset to make analysis easier downstream.

age_group

Using pandas `qcut()`, customers were divided into four equal-sized age groups: Young Adult, Adult, Middle-aged, and Senior. This makes it much easier to slice marketing data by age without having to deal with continuous age bins manually in every query.

Age Group	Approx. Age Range	Rationale
Young Adult	18 – 30	Entry-level buyers, trend-driven
Adult	31 – 44	Stable income, value-conscious
Middle-aged	45 – 57	Higher spend potential
Senior	58 – 70	Brand-loyal, less price-sensitive

Table 6: Age group definitions

purchase_frequency_days

The original `frequency_of_purchases` column was text (e.g., 'Fortnightly', 'Every 3 Months'). A numeric column was added by mapping each label to an approximate number of days, making it possible to sort, filter, and aggregate by purchase cadence in SQL.

Frequency Label	Days
Weekly	7
Bi-Weekly / Fortnightly	14
Monthly	30
Quarterly / Every 3 Months	90
Annually	365

Table 7: Frequency-to-days mapping

6. SQL Integration

After cleaning, the dataset was loaded into a PostgreSQL database using SQLAlchemy. The connection was set up from a Kaggle notebook, which created an interesting infrastructure challenge: Kaggle runs on

cloud servers that can't reach a local machine's network directly.

The solution was to use **Bore**, an open-source tunneling tool, to expose the local PostgreSQL instance (port 5432) through a public URL. Running `bore local 5432 --to bore.pub` created a temporary bridge that the Kaggle notebook could reach.

A second issue surfaced during upload: the tunnel would time out when trying to send 3,900 rows in a single batch. This was fixed by adding `chunksizes=500` to the `to_sql()` call, which sent data in smaller batches and kept the connection stable throughout.

Technical note: The tunnel address and port change every session since Bore assigns them dynamically. The connection credentials need to be updated each time the tunnel is restarted. For a production setup, a persistent DB server would remove this friction entirely.

7. Business Recommendations

R1: Grow the Subscription Base

Only 27% of customers subscribe, yet average previous purchases sit at 25 per customer. A lot of repeat buyers simply aren't subscribing. A targeted re-engagement campaign offering a first-month discount or exclusive perks to high-frequency non-subscribers could close that gap.

R2: Rethink the Discount Strategy

43% of purchases use a promo code. That's high enough to suggest some customers may be holding off on full-price purchases. Running an A/B test on discount-free campaigns for high-frequency buyers would reveal whether discounts are actually driving incremental purchases or just eroding margin.

R3: Capitalize on Seasonal Consistency

Purchase volumes are almost identical across all four seasons. Rather than treating this as a neutral fact, the business could lean into seasonal campaigns to create deliberate peaks — especially for Outerwear in winter and Footwear in spring, where category-season alignment is natural.

R4: Segment Marketing by Age Group

The `age_group` feature makes it easy to run targeted campaigns. Young Adults (18–30) and Seniors (58–70) likely respond to very different messaging. Testing age-segmented campaigns before rolling out broad promotions would improve ROI and reduce blanket discount dependency.

R5: Simplify Payment Options Strategically

Six payment methods with nearly equal usage means significant backend complexity for marginal differentiation. Analyzing which methods correlate with higher average order value or repeat purchases would help prioritize where to invest in payment UX improvements.

8. Conclusion

This analysis covered the full pipeline from raw CSV to structured insights: data cleaning, feature engineering, SQL ingestion, and business interpretation. The dataset is clean and well-structured, which made the work straightforward. The harder questions are the business ones.

A few things stood out. Purchase behavior is remarkably uniform — categories, seasons, payment methods, and shipping preferences are all close to evenly distributed. That kind of flatness is either a sign of a genuinely diverse customer base, or a signal that there are patterns the current data isn't capturing (time series, geographic clustering, etc.).

The subscription and discount dynamics are the most actionable findings here. A 27% subscription rate among customers who average 25 previous purchases each is a real gap. And a 43% promo-code reliance deserves scrutiny before it becomes a habit that's hard to break.

The next logical step is building the Power BI dashboard to make these findings accessible to non-technical stakeholders — and running SQL queries against the PostgreSQL database to dig into customer segmentation at a more granular level.

End of Report | All figures derived from the customer_shopping_behavior.csv dataset